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Lecture 04A

Before-Tutorial-Session

Tutorial 3

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HIDDEN

Let $S = \{v_1, v_2, \dots, v_n\}$.

The set S is linearly independent if

$$(*) \quad \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n = \text{BLANK1}$$

Implies $\lambda_1 = \lambda_2 = \dots = \lambda_n = \text{BLANK2}$

vector

zero vector

number zero

number

Fill in the blanks. The equation (*) has exactly one solⁿ
iff S is linearly independent

What is your tutor's name?

We will solve:

Question 2. Prove the following statements.

- (a) $\{v\}$ is linearly independent if and only if v is not a zero vector.
- (b) $\{u, v\}$ is linearly independent if and only if u is not a scalar multiple of v and v is not a scalar multiple of u .
- (c) If $0 \in S$, then S is not linearly independent.

consider a set S with one
vector x
 S is linearly independent
iff $x \neq 0$

If S has a zero vector, then S is
not linearly independent
▽

We go through 2(a) and 2(c)

Question 2. Prove the following statements.

(a) $\{v\}$ is linearly independent if and only if v is not a zero vector.

Is Albert's proof that "If v is nonzero, then $\{v\}$ is linearly independent" correct?

Albert's Solution: Let $v = (1048576, 0)$.

Then we consider

$$\lambda(1048576, 0) = (0, 0) \quad (*)$$

This implies that $1048576\lambda = 0$ and so, $\lambda = 0$

Therefore, $\{v\}$ is linearly independent.

— Not correct - we have to prove for ALL possibilities for v

Question 2. Prove the following statements.

(a) $\{v\}$ is linearly independent if and only if v is not a zero vector.

Proof that "If v is nonzero, then $\{v\}$ is linearly independent"

Suppose that $v \neq 0$

$$\lambda v = 0 \quad \text{--- (0)}$$

Then either $\lambda = 0$ OR $v = 0$ (not possible because $v \neq 0$)

$$\therefore \lambda = 0$$

$\Rightarrow \{v\}$ is linearly independent

Question 2. Prove the following statements.

(a) $\{v\}$ is linearly independent if and only if v is not a zero vector.

Consider the statement:

"If $\{v\}$ is linearly independent, then v is nonzero."

→ if $\neg q$ then $\neg p$

What is the contrapositive of the above statement?

1. If $\{v\}$ is NOT linearly independent, then $u = 0$.

✓ 2. If $v = 0$, then $\{v\}$ is NOT linearly independent.

Proof that "If $\{v\}$ is linearly independent, then v is nonzero". WTS $\{v\}$ is not L.I.

Contrapositive : $v = 0$

Consider $\lambda v = 0$ — ($\star$)

~~Either $\lambda = 0$ OR $v = 0$~~

Choose $\lambda = 1048576$. This is a non trivial solution to ($\star$)

$\therefore \{v\}$ is not linearly independent

$\star$ As long as you find nontrivial soln, it is not L.I..

Question 2. Prove the following statements.

(c) If $\mathbf{0} \in S$, then S is not linearly independent.

↪ not linearly independent

If $\mathbf{0} \in S$, let us write $S = \{\mathbf{0}, \mathbf{v}_1, \dots, \mathbf{v}_t\}$. Consider the equation

$$\lambda \mathbf{0} + \mu_1 \mathbf{v}_1 + \dots + \mu_t \mathbf{v}_t = \mathbf{0} \quad (*)$$

Which of the following is definitely a non-trivial solution to $(*)$?

~~1.~~ $\lambda = \mu_1 = \dots = \mu_t = 1048576$ LHS may not be 0

~~2.~~ $\lambda = \mu_1 = \dots = \mu_t = 0$ (trivial solⁿ)

~~3.~~ $\lambda = 1048576, \mu_1 = \dots = \mu_t = 0$ LHS must be 0

Since we have a nontrivial solution, S must be linearly independent

$$\dim(\mathcal{V}) = d$$

Let $\mathcal{V}$ be a d -dimensional vector space.

Consider the following statements.

- a) S is a basis for $\mathcal{V}$ if S is linearly independent and $\text{span}(S) = \mathcal{V}$.
- b) S is a basis for $\mathcal{V}$ if $|S| = d$ and $\text{span}(S) = \mathcal{V}$.
- c) S is a basis for $\mathcal{V}$ if $|S| = d$ and S is linearly independent.

Which of these statements is/are true?

- 1. Only (a) is true
- 2. All (a), (b) and (c) are true

Theorem (Criteria for a Basis).

Let $\mathcal{V}$ be a vector space with $\dim(\mathcal{V}) = d$.

Then a set S is a basis for $\mathcal{V}$ if two out of the following three holds.

(I) $|S| = d$,

(II) $\text{span}(S) = \mathcal{V}$,

(III) the set S is linearly independent.

If S is a basis, all three conditions hold.

(II) $\nrightarrow$ (III) is definition for Basis

! Prof's preference to check : (I) $\nrightarrow$ (II)

Relies on the fact
that we know: $\dim(\mathcal{V})$
This is true for some
standard vector spaces

Standard Bases

$[E_{ij} : 1 \leq i \leq m, 1 \leq j \leq n]$ is the standard basis for $\mathcal{M}_{m \times n}(\mathbb{R})$.

Here, E_{ij} is the matrix with one at (i, j) -th position and zero elsewhere.

What is the dimension of $\mathcal{M}_{m \times n}(\mathbb{R})$?

1. $m + n + 1$
- ✓ 2. mn

$$\dim(\mathcal{M}_{m \times n}) = mn$$

suppose $m=2, n=3$

vectors are of the form:

$$\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$$

standard basis

$$\left[\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right]$$

Standard Bases

$[1, x, \dots, x^d]$ is the standard basis for $\mathcal{P}_d(\mathbb{R})$.

What is the dimension of $\mathcal{P}_d(\mathbb{R})$?

1. d
- ✓ 2. $d + 1$

$$\dim(\mathcal{P}_d) = d + 1$$

Suppose $d = 3$
polys are of the form
 $a + bx + cx^2 + dx^3$

Standard Basis : $[1, x, x^2, x^3]$

We will solve:

Question 8. Determine whether the following list $\mathcal{S}$ is a basis for $\mathcal{V}$.

(a) $\mathcal{S} = [(-1, 3, 1, 0), (2, -4, 0, -3), (-3, 0, 8, 2), (0, 0, 1, -2)], \quad \mathcal{V} = \mathbb{R}^4.$

(b) $\mathcal{S} = [(3, 1, -1, 0), (2, 0, -3, 4), (1, 0, 0, 5), (3, 2, 5, 1), (2, 0, 1, 9)], \quad \mathcal{V} = \mathbb{R}^4.$

(c) $\mathcal{S} = [1 + 2x - x^2, 4 - 2x + x^2, -1 + 18x - 9x^2], \quad \mathcal{V} = \mathcal{P}_2(\mathbb{R}).$

(d) $\mathcal{S} = [\pi, x - \sqrt{2}x^3, 1102 - 3x^2, -6x, x^2 + 2019x^3], \quad \mathcal{V} = \mathcal{P}_3(\mathbb{R}).$

$$\dim(\mathcal{P}_2(\mathbb{R})) = 3$$

Solution Plan

We check two things:

- $\mathcal{S}$ has the correct size.
- either $\text{span}(\mathcal{S}) = \mathcal{V}$ or $\mathcal{S}$ is linearly independent

↓
prof's preference

8(c)

Set $\mathcal{S} = [1 + 2x - x^2, 4 - 2x + x^2, -1 + 18x - 9x^2]$.Set $\mathcal{V} = \mathcal{P}_2(\mathbb{R})$.Is $\mathcal{S}$ a basis for $\mathcal{V}$?

1. Yes
2. No

$$\dim(\mathcal{P}_2(\mathbb{R})) = 3$$

- $|\mathcal{S}|$ correct? Yes- $\mathcal{S}$ is L.I.? NoDoes it have a non-trivial solⁿ?

$$\begin{aligned} & \lambda_1(1 + 2x - x^2) \\ & + \lambda_2(4 - 2x + x^2) \\ & + \lambda_3(-1 + 18x - 9x^2) \\ & = 0 \end{aligned} \longrightarrow$$

$$\lambda_1 + 4\lambda_2 - \lambda_3 = 0$$

$$2\lambda_1 - 2\lambda_2 + 18\lambda_3 = 0$$

$$-\lambda_1 + \lambda_2 - 9\lambda_3 = 0$$

Yes.

$$\lambda_1 = -7$$

$$\lambda_2 = 2$$

$$\lambda_3 = 1$$

Set $\mathcal{S} = [\pi, x - \sqrt{2}x^3, 1102 - 3x^2, -6x, x^2 + 2019x^3]$.

Set $\mathcal{V} = \mathcal{P}_3(\mathbb{R})$.

Is $\mathcal{S}$ a basis for $\mathcal{V}$?

1. Yes
- ✓ 2. No

$$\dim(\mathcal{V}) = \dim(\mathcal{P}_3(\mathbb{R})) = 4$$

- $|\mathcal{S}|$ correct? No. $|\mathcal{S}| = 5$ here

- $\mathcal{S}$ is L.I.?

We will solve:

Question 5. Consider the list $\mathcal{B} = [u_1, u_2, u_3, u_4] \subseteq \mathbb{R}^4$, where

$$u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad u_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad u_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \quad u_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

- (a) Explain why $\mathcal{B}$ is a basis for $\mathbb{R}^4$.
- (b) Find the unique representation of an arbitrary vector $v \in \mathbb{R}^4$ as a linear combination of vectors in $\mathcal{B}$.
- ✓ (c) Find the coordinate vector $[e_1]_{\mathcal{B}}$ with respect to the basis $\mathcal{B}$.

We go through 5(c)

Let $\mathcal{B} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n]$ be a basis for $\mathcal{V}$ and let $\mathbf{u} \in \mathcal{V}$.

If $\mathbf{u} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n$, the coordinate vector of $\mathbf{u}$ with respect to $\mathcal{B}$ is $(\lambda_1, \lambda_2, \dots, \lambda_n)^T$. We write this as $[\mathbf{u}]_{\mathcal{B}}$.

Example. $\mathcal{B} = [(1,1,1,1)^T, (0,1,1,1)^T, (0,0,1,1)^T, (0,0,0,1)^T]$ is a basis for $\mathbb{R}^4$.

Find the coordinate vector of $\mathbf{e}_1$ with respect to $\mathcal{B}$.

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \lambda_1 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \lambda_2 \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \lambda_3 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} + \lambda_4 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \text{solve linear equations to find } \lambda_1, \lambda_2, \lambda_3, \lambda_4$$

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} \leftarrow \text{coordinate vector of } \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ wrt } \mathcal{B} \\ = [\mathbf{e}_1]_{\mathcal{B}}$$

Theorem (Unique Representation). Let $S = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n]$.

S is a basis for $\mathcal{V}$ if and only if every $\mathbf{u} \in \mathcal{V}$ can be uniquely written in the form

$$\mathbf{u} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n \text{ for some } \lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}.$$

(Basis) Let $\mathcal{V}$ be a vector space.

The set $S \subseteq \mathcal{V}$ is a basis for $\mathcal{V}$ if

- $\text{span}(S) = \mathcal{V}$, and
- the set S is linearly independent.

— Detailed proof is in colab

* This implies $[\mathbf{u}]_B$ is
unique

Set $\mathcal{B} = [(1,1,1,1)^T, (0,1,1,1)^T, (0,0,1,1)^T, (0,0,0,1)^T]$.

Then $\mathcal{B}$ is a basis for $\mathbb{R}^4$

Find the coordinate vector of $\mathbf{e}_1$ with respect to $\mathcal{B}$.

✓ 1. $[\mathbf{e}_1]_{\mathcal{B}} = (1, -1, 0, 0)^T$

2. $[\mathbf{e}_1]_{\mathcal{B}} = (1, 0, 0, 0)^T$

$$[\mathbf{e}_1]_{\mathcal{B}}$$